Spectral–Spatial Attention Network for Hyperspectral Image Classification

Hao Sun, Xiangtao Zheng, *Member, IEEE*, Xiaoqiang Lu, *Senior Member, IEEE*, and Siyuan Wu

***Abstract*— Hyperspectral image (HSI) classification aims to assign each hyperspectral pixel with a proper land-cover label. Recently, convolutional neural networks (CNNs) have shown superior performance. To identify the land-cover label, CNN-based methods exploit the adjacent pixels as an input HSI cube, which simultaneously contains spectral signatures and spatial information. However, at the edge of each land-cover area, an HSI cube often contains several pixels whose land-cover labels are different from that of the center pixel. These pixels, named interfering pixels, will weaken the discrimination of spectral–spatial features and reduce classification accuracy. In this article, a spectral–spatial attention network (SSAN) is proposed to capture discriminative spectral–spatial features from attention areas of HSI cubes. First, a simple spectral–spatial network (SSN) is built to extract spectral–spatial features from HSI cubes. The SSN is composed of a spectral module and a spatial module. Each module consists of only a few 3-D convolution and activation operations, which make the proposed method easy to converge with a small number of training samples. Second, an attention module is introduced to suppress the effects of interfering pixels. The attention module is embedded into the SSN to obtain the SSAN. The experiments on several public HSI databases demonstrate that the proposed SSAN outperforms several state-of-the-art methods.**

***Index Terms*— Attention, convolutional neural network (CNN), hyperspectral image (HSI) classification, spectral–spatial feature extraction.**

I. Introduction

With the rapid development of remote sensing hyperspectral imaging technology, it is easy to capture hyperspectral images (HSIs) for Earth observation. HSI classification attempts to identify the category for each hyperspectral pixel [1]–[3], which is helpful to remote sensing applications, such as geologic exploration, urban planning, and forest monitoring. Hyperspectral image (HSI) is a 3-D data, which is composed of 1-D spectral signatures and 2-D spatial information. The spectral signatures (pixels in HSI) consist of hundreds of narrow spectral bands, which contain plentiful physical properties of land covers. The spatial information contains the detailed spatial distribution of land covers. Both spectral signatures and spatial information are effective clues for HSI classification.

Recently, deep learning (DL) has attracted wide attention in the remote sensing community [4]–[6]. A large number of DL-based methods have been proposed for HSI classification [7]. The DL-based methods [8], [9] directly learn high-level abstract features from HSIs, which show better performance than the traditional handcrafted feature-based methods (e.g., morphological profiles [10], [11] and attribute profiles [12]). According to whether the spatial information is exploited, DL-based methods can be roughly divided into the spectral DL-based methods and the spectral–spatial DL-based methods. The spectral DL-based methods focus on making full use of the spectral signatures of each pixel for HSI classification. The deep belief network [13], stacked autoencoder [14], and recurrent neural network (RNN) [15] are usually exploited to extract spectral features from spectral signatures. However, due to the spectral variability caused by atmospheric effects and instrument noises [16], [17], the classification results of spectral DL-based methods are still unsatisfactory. The spectral–spatial DL-based methods attempt to utilize the adjacent pixels to complement the spectral signatures for HSI classification. Based on the Tobler's First Law of Geography, adjacent pixels in HSIs are assumed to share the same land-cover label [1]. Based on this assumption, adjacent pixels are complementary to spectral signatures. The spectral–spatial DL-based methods can extract spectral–spatial features from the adjacent pixels and have shown outstanding performance in the HSI classification task.

The convolutional neural network (CNN) [18] is one of the most popular deep frameworks in spectral–spatial DL-based methods for extracting spectral–spatial features from the adjacent pixels. Taking the Indian Pines database as an example, the overall architecture of exploiting a CNN for HSI classification is briefly introduced in Fig. 1. First, as shown in Fig. 1(a), HSI cubes with a fixed spatial size are cropped from the original Indian Pines database to satisfy the input characteristics

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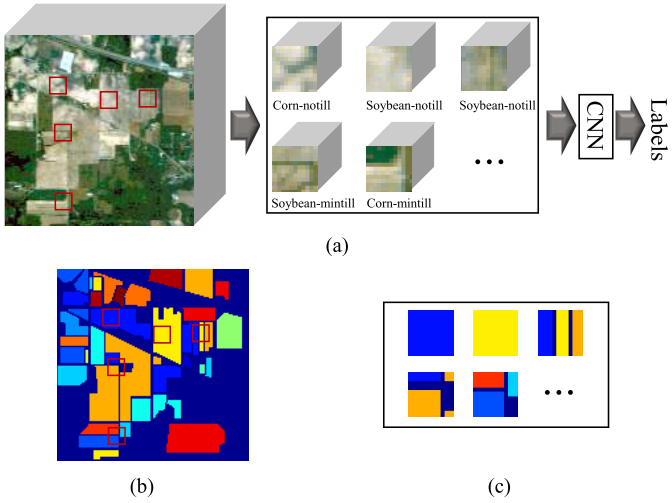


Fig. 1. (a) In CNN-based HSI classification methods, hyperspectral data are first cropped into plenty of HSI cubes for extracting spectral–spatial features. However, as shown in (b) and (c), at the edge of each land-cover area, cropped HSI cubes usually contain several pixels whose land-cover labels are different from that of the center pixel. These pixels will interfere with the discrimination of spectral–spatial features. (a) Overall architecture of using CNN for HSI classification. (b) Land-cover map of Indian pines. (c) Land-cover map of cropped cubes

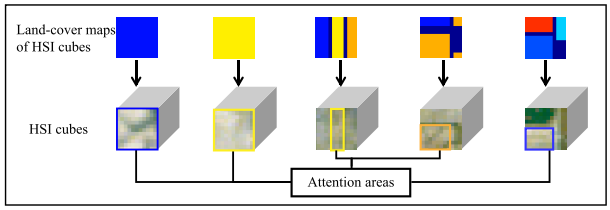


Fig. 2. Land-cover map of each HSI cube is employed to help illustrate the attention areas in HSI cubes. In the land-cover map, the same color represents the same land-cover category. In an HSI cube, pixels whose land-cover labels are the same as the land-cover label of the center pixel form attention areas.

of CNN. Second, a CNN is exploited to extract spectral–spatial features from HSI cubes. Based on the assumption that the adjacent pixels are highly likely to have the same land-cover label, the HSI cube can be appropriately identified as the label of center pixel. Finally, land-cover labels of hyperspectral pixels are inferred based on the spectral–spatial features. However, as shown in Fig. 1(b) and (c), at the edge of each land-cover area, cropped HSI cubes usually contain several pixels whose land-cover labels are different from the land-cover label of center pixel. These pixels with land-cover labels different from the center pixel are named interfering pixels. When HSI cubes are fed into a CNN, interference pixels will interfere with the extraction of spectral–spatial features. A natural idea to suppress the effects of interfering pixels in HSI cubes is to extract spectral–spatial features directly from pixels whose land-cover labels are the same as that of center pixel. As shown in Fig. 2, in an HSI cube, the pixels whose land-cover labels are the same as the land-cover label of the center pixel form attention areas. The motivation of this article is to capture spectral–spatial features from attention areas of HSI cubes for HSI classification.

In this article, a spectral–spatial attention network (SSAN) is proposed to extract discriminative spectral–spatial features from attention areas for HSI classification. The proposed SSAN is composed of a simple spectral–spatial network (SSN) and an attention module. The SSN is proposed to extract spectral–spatial features from HSI cubes, which consists of a spectral module and a spatial module in series. First, the spectral module is utilized to extract spectral features of each pixel in HSI cubes. Then, the spatial module is added following the spectral module to extract spectral–spatial features. Both the spectral module and the spatial module contain only a few simple 3-D convolution and activation operations, which make the SSN easy to converge with a small number of training samples. The attention module is exploited to extract discriminative spectral–spatial features from attention areas of HSI cubes. The attention module leverages the correlation between hyperspectral pixels within an HSI cube to capture attention areas. The attention modules are added following the spectral module and each convolutional layer of the spatial module to obtain the end-to-end SSAN for HSI classification.

The contributions of this article are as follows.

1) A simple SSN is proposed to extract the spectral–spatial features from HSI cubes. The proposed SSN contains only a few simple 3-D convolution and activation operations, which make SSN easy to converge.
2) An attention module is proposed to extract the discriminative features from attention areas of HSI cubes. The attention module can be embedded anywhere in the SSN for HSI classification.
3) An SSAN is proposed to extract discriminative spectral–spatial features for HSI classification. The proposed SSAN integrates the spectral–spatial feature extraction and the attention area extraction into a unified network for the end-to-end training.

The remaining sections are organized as follows. First, the works related to HSI classification are reviewed in Section II. Then, the proposed SSAN is introduced in Section III. Next, experimental evaluations of SSAN are reported in Section IV. Finally, Section V concludes this article.

II. Related Work

A. DL-Based HSI Classification Methods

HSI classification, which aims to assign a land-cover label to each pixel, has recently attracted considerable attention [19]. In this section, the DL-based HSI classification methods are reviewed. The DL-based methods can be summarized into the spectral DL-based methods and the spectral–spatial DL-based methods.

The spectral DL-based methods infer pixel labels by only using the spectral signatures. The 1-D spectral signatures are fed to the DL framework for capturing deep spectral features. Chen *et al.* [14] and Lin *et al.* [20] exploited the unsupervised stacked autoencoder to extract deep spectral features and employed SVM for classification. Chen *et al.* [13] utilized the deep belief network to capture the spectral features from

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1-D spectral signatures. Hu *et al.* [21] proposed a 1-D CNN to classify HSI in the supervised manner. Mou *et al.* [15] treated the spectral signature as a sequence and employed the RNN to analyze the hyperspectral sequential data. Wu and Prasad [22] utilized the 1-D CNN in combination with an RNN to capture the contextual information from spectral bands for HSI classification. Zhan *et al.* [23] proposed a novel generative adversarial network (GAN) to alleviate the problem of insufficient labeled HSI pixels. The HSI pixels were fed to the discriminator of the proposed GAN to obtain the classification results. However, the spectral signatures in the actual imaging process are easily disturbed by the atmospheric effects, instrument noises, and incident illumination [1], [16]. It is difficult to obtain satisfactory results by using only spectral signals for classification.

Different from the abovementioned methods, the spectral–spatial DL-based methods not only consider the spectral signatures but also exploit the spatial information from HSI classification. In fact, the HSI is a natural 3-D data, one of which is the spectral signature, and the other two are the spatial structure information. Early spectral–spatial DL-based methods try to employ the flattening operation to make use of spatial information to complement spectral information [13], [14], [20], [24]. First, the HSI cube is flattened to a 1-D vector. Then, the flattened vector is fed to original spectral DL-based methods. However, the spatial structure information has been destroyed in the flattening operation.

Recently, the CNN has become a popular framework for extracting spectral–spatial features, in which the spatial structure information can be preserved. Many works [9], [18], [25], [26] focus on designing novel deep spectral–spatial convolutional networks to extract features directly from HSI. Li *et al.* [18] exploited a CNN to extract spectral–spatial features directly from raw HSI cubes without any preprocessing operations. Chen *et al.* [25] designed a 3-D CNN to extract spectral–spatial features directly from the HSI cubes. Mou *et al.* [26] proposed an unsupervised residual conv–deconv network to capture spectral–spatial features from unlabeled 3-D HSI cubes. Zhong *et al.* [9] proposed a spectral–spatial residual network to capture spectral features and spatial features in sequential residual convolutional blocks. Zhu *et al.* [27] proposed a deformable CNN to extract deformable features from neighboring similar pixels. Hao *et al.* [28] proposed a two-stream deep architecture by combining a CNN and a stacked denoising autoencoder. Both the spectral pixels and spatial patches are employed to predict classification results. Fang *et al.* [29] proposed a squeeze multibias network (SMBN) that decouples convolutional feature maps into multiple response maps first. Then, proper response maps are selected for HSI classification. Song *et al.* [30] built a deep feature fusion network (DFFN) to take advantage of the complementary of different hierarchical layers. Ma *et al.* [31] designed a deconvolution network with a band selection layer to avoid overfitting and leverage the appearance information from shallow convolutional layers for HSI classification. Zhang *et al.* [32] proposed a dual-channel CNN to extract the spectral feature and spatial feature, respectively. Santara *et al.* [33] proposed a band-specific feature learning network to reduce the weight parameters and training time. Lee and Kwon [34] exploited the multiscale convolutional filters to build a contextual CNN, which can utilize local contextual information for classification. Paoletti *et al.* [35] proposed a spectral–spatial capsule network to reduce the complexity of convolutional networks. In addition to designing deep networks, some recent works [36]–[40] explore special techniques to improve the discrimination of spectral–spatial features. Aptoula *et al.* [36] exploited the attribute profiles [41] to provide the spectral and geometric properties of HSI for the CNN. Li *et al.* [37] proposed a data augmentation method and a pixel-pair convolutional network to mitigate the problem of insufficient labeled samples. Chen *et al.* [38] proposed to combine a CNN with Gabor filters to strengthen texture and edge information. Mei *et al.* [39] utilized a CNN to learn sensor-specific features for HSI. Li *et al.* [42] leveraged the principal component analysis (PCA) and a fully convolutional network to enhance spectral–spatial features. Hao *et al.* [40] exploited a super-resolution network to increase the number of labeled HSI pixels and utilized an unsupervised triplet-pipeline CNN to extract deep discriminative features. Cao *et al.* [43] first exploited the CNN to learn the posterior distribution of land-cover categories and then employed the Markov random fields (MRFs) to smooth classification results. He *et al.* [44] proposed a multiscale covariance map to extract the features from the convolutional features of CNN. Zhang *et al.* [45] proposed a diverse region-based CNN to jointly learn contextual features from six diverse HSI cubes. In summary, spectral–spatial DL-based methods have achieved good results in the HSI classification and have become the mainstream methods of HSI classification.

Although there are many kinds of spectral–spatial DL-based methods, the CNN is still an indispensable part of those methods to capture spectral–spatial features from the adjacent pixels. Due to the particularity of the CNN structure, its input must be a cube. Almost all spectral–spatial DL-based methods are based on the assumption that adjacent pixels have the same land-cover label to crop HSI cubes from original hyperspectral data. The cropped HSI cube is labeled with the label of its center pixel. However, at the edge of each land-cover area, the cropped HSI cube usually contains interfering pixels, whose land-cover labels are different from the land-cover label of the current HSI cube. When the HSI cubes are input into CNN, the interference pixels will interfere with the extraction of spectral–spatial features. To mitigate this problem, an SSAN is proposed to extract discriminative spectral–spatial features from the attention areas of HSI cubes.

B. Attention Mechanism

The attention mechanism is inspired by the human visual system to capture key areas from images for classification [46], [47]. Recently, the attention mechanism has received increasing attention in the field of remote sensing. Early works [48], [49] employ the saliency detection that is similar to the attention mechanism to capture salient areas. Zhang *et al.* [48] employed a saliency detection method as the sampling strategy to extract representative salient areas.

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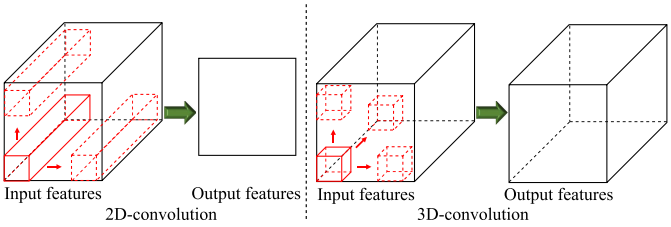


Fig. 3. Illustration of 2-D convolution and 3-D convolution.

Diao *et al.* [49] utilized the norm gradient map as the saliency map. However, salient areas are not always important for classification [46]. Recently, Wang *et al.* [46] proposed to utilize several long short-term memory (LSTM) networks to recurrently extract attention features from deep convolutional features of CNN. However, the LSTM is a complex neural network with plenty of weight parameters. Combining LSTM and CNN to extract attention features makes the structure of attention recurrent convolutional network [46] too complex to flexible. Haut *et al.* [47] proposed a two-branch attention module, where one branch is utilized to generate an attention mask and the other branch is utilized to extract convolutional features. Then, attention features are obtained by multiplying the attention mask with the convolutional features. Different from previous methods [46], [47], an attention module that incorporates attention feature extraction into a simple component is proposed in this article. The proposed attention module exploits the correlation between the hyperspectral pixels within an HSI cube to capture attention features. The proposed attention module can be flexibly embedded into the CNN to extract discriminative spectral–spatial features for HSI classification.

III. SSAN

A. CNN for HSI Classification

The CNN is a very popular architecture for extracting spectral–spatial features in the spectral–spatial DL-based methods. This article also employs the CNN for HSI classification. This section briefly describes the process of HSI classification with a CNN.

First, the notations related to HSI classification are defined. Let $X \in \mathbb{R}^{h\times w\times c}$ represent the HSI, where h and w denote the spatial height and width, respectively, and c denotes the band number of spectral signature. The CNN-based methods attempt to explore the spectral–spatial information from the adjacent pixels for HSI classification. In order to satisfy the common design of CNN whose inputs are square images, a square box with fixed length is usually set around the pixel to extract the adjacent pixels. Adjacent pixels in the square box form an HSI cube. Therefore, in the CNN-based methods, the HSI X is first cropped into a total of hw HSI cubes, where each pixel of X corresponds to an HSI cube. The set of HSI cubes is represented as $\mathcal{X} = \{X_1, X_2, \ldots, X_{hw}\}$, where $X_i \in \mathbb{R}^{s\times s\times c}$ and $s \times s$ is the spatial size of HSI cubes. Second, the land-cover label of HSI cube X_i is determined by its central pixel. The land-cover label set that corresponds to $\mathcal{X}$ is represented as $\mathcal{Y} = \{Y_1, Y_2, \ldots, Y_{hw}\}$, where $Y_i \in \{1, 2, \ldots, n\}$, with n the number of land-cover categories. Then, $(\mathcal{X}, \mathcal{Y})$ can be utilized to represent the sample set. A certain number of samples are randomly selected from each land-cover category to form training samples, and the rest is used as testing samples.

To fully explore the spectral–spatial information of $(\mathcal{X}, \mathcal{Y})$, a variety of deep convolutional networks have been proposed, e.g., deep pyramidal residual networks [50], DFFNs [30], and spectral–spatial residual networks [9]. The basic components of CNNs are very simple, such as convolution, activation, fully connection, batch normalization, and skip connection. Previous works usually focus on designing deep and complicated networks with these basic components for HSI classification. Little attention is devoted to extracting features from attention areas of HSI cubes. In this article, an SSAN is proposed to extract the spectral–spatial features from attention areas. The SSAN is composed of a spectral module, a spatial module, and several attention modules. Finally, the softmax is employed for classifying. The details of SSAN are introduced as follows.

B. Spectral–Spatial Convolutional Network

Similar to the previous works [9], a simple spectral–spatial network (SSN) is built to extract deep spectral–spatial features from HSI cubes first. The proposed SSN is composed of two modules: the spectral module and the spatial module. The spectral module and the spatial module are connected in series. The spectral module is exploited to extract spectral features of each pixel in HSI cubes. Then, the spatial module is utilized to capture spectral–spatial features from the spectral features of the spectral module. In this article, the 3-D convolution is employed as a basic convolutional operation to build the network. The difference between 3-D convolution and 2-D convolution is shown in Fig. 3. In a 2-D convolutional layer, the convolutional kernels only slide in the spatial dimension to calculate the convolution, while the convolutional kernels slide in both the spatial dimension and the channel dimension to calculate the convolution in a 3-D convolutional layer. The spectral dimension of HSI contains the detailed physical properties of land covers. The 3-D convolution is a proper operation to explore spectral information of HSI for classification.

1) Spectral Module: HSIs have a large number of spectral bands. Early works [13], [30] usually employ the unsupervised PCA to extract the spectral features. However, the unsupervised PCA cannot fully extract features that are discriminative to HSI classification [51]. In this article, a spectral module is utilized to extract spectral features directly from each pixel of HSI cubes. The spectral module can be combined with the spatial module for joint training in an end-to-end manner.

The spectral module is simply composed of several 3-D convolutional layers with the rectified linear unit (ReLU), where the size of convolutional kernels is $1 \times 1 \times 7$ (height $\times$ width $\times$ channel), the stride of convolution is $1 \times 1 \times 1$, and the number of convolutional kernels is 32. In the spectral module, the convolution is conducted only on the channel of HSI cubes.

2) Spatial Module: To explore the spatial information of HSI cubes, a spatial module is proposed. The spatial module also consists of several convolutional layers with the ReLU. However, different from the spectral module, the size of

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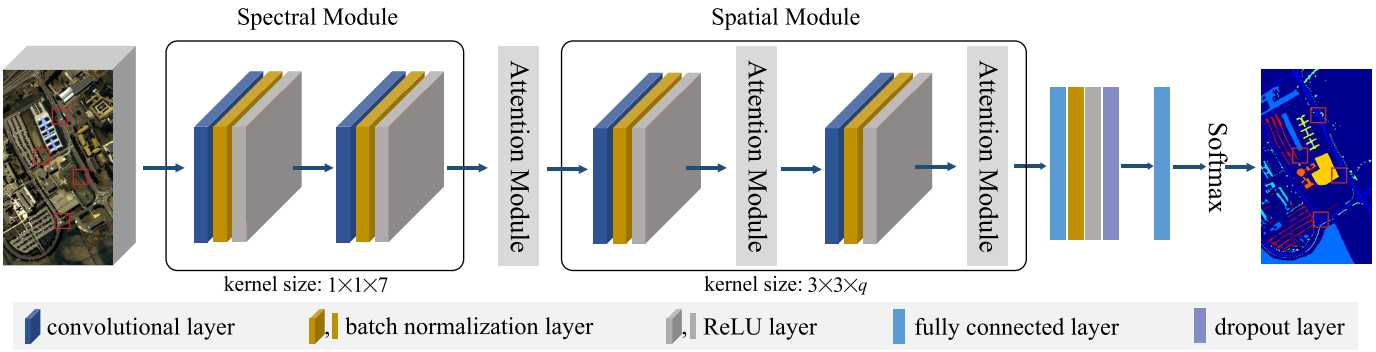


Fig. 4. Architecture of the proposed SSAN. The Pavia University database is utilized as an example to illustrate the architecture. The spectral module is simply composed of two convolutional layers, two batch normalization layers, and two ReLU layers. The spatial module is composed of two convolutional layers, two batch normalization layers, two ReLU layers, and two attention modules. In the spatial module, the kernel size is $3 \times 3 \times q$, where q is the channel number of input features. The padding operation is employed in each convolutional layer to keep the spatial size of convolutional features always 7×7. Finally, two fully connected layers and a softmax layer are utilized to classify.

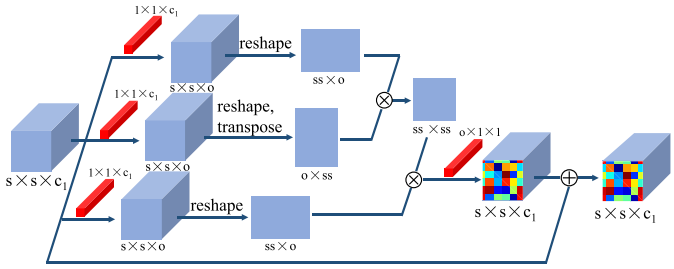


Fig. 5. Structure of the attention module. $\otimes$ represents the matrix multiplication. $\oplus$ represents the matrix addition.

convolutional kernels in the spatial module is $3 \times 3 \times q$, where q is the channel number of inputs of the current convolutional layer. The stride of convolution is also $1 \times 1 \times 1$. The number of convolutional kernels is 64. In the spatial module, the convolution is conducted both on the spatial and spectral dimensions.

The architecture of the spectral module is similar to that of the spatial module. The difference between the spectral module and the spatial module is the size of convolutional kernels. The convolutional kernels with the spatial size 1×1 make the spectral module focus on extracting spectral features. The convolutional kernels with the spatial size 3×3 allow the spatial module to extract complementary spatial information from the adjacent pixels. By connecting the spectral module and the spatial module in series to form the SSN, the spectral–spatial features can be extracted from HSI cubes. In addition, the padding operation is exploited in the convolutional layer to maintain the spatial size of features.

C. Attention Module

In the proposed network, the motivation of the attention module is to suppress the effects of interfering pixels. In order to extract attention features of HSI cubes, the self-attention mechanism [52] is introduced to construct the attention module. The attention module exploits the correlation between the hyperspectral pixels within an HSI cube to obtain attention areas and capture spectral–spatial attention features. The attention module is a flexible component that can be embedded in the SSN. As shown in Fig. 4, in this article, attention modules are added following the spectral module and each convolutional layer of the spatial module. The spectral module of SSN only focuses on extracting spectral features, and the attention module will introduce the spatial information. Therefore, the attention module is not embedded in the spectral module.

The attention module following the spectral module is utilized as an example for the detailed introductions. The structure of the attention module is shown in Fig. 5. Define $X_k^l \in \mathbb{R}^{s \times s \times c_1}$ to represent the spectral feature of the HSI cube X_k, where $s \times s$ is the spatial size and c_1 is the channel number. First, the 3-D convolutional layer with $1 \times 1 \times c_1$ kernels is employed to transform X_k^l into $f(X_k^l) \in \mathbb{R}^{s \times s \times o}$, $g(X_k^l) \in \mathbb{R}^{s \times s \times o}$, and $h(X_k^l) \in \mathbb{R}^{s \times s \times o}$ for calculating the attention, where o represents the number of $1 \times 1 \times c_1$ convolutional kernels. The details of $f(X_k^l)$ can be formulated as follows. First, X_k^l is convoluted with kernel W_f^t to obtain $f^t(X_k^l) \in \mathbb{R}^{s \times s}$. Then, the features $f^t(X_k^l)$ $(t = 1, 2, \ldots, o)$ are concatenated by the channel to obtain $f(X_k^l)$

$$f^t(X_k^l) = \sigma\left(W_f^t * X_k^l + b_f^t\right), \quad t = 1, 2, \ldots, o \tag{1}$$

$$f(X_k^l) = \left[f^1(X_k^l), f^2(X_k^l), \ldots, f^o(X_k^l)\right] \tag{2}$$

where $W_f^t \in \mathbb{R}^{1 \times 1 \times c_1}$ represents the weight parameter, $b_f^t \in \mathbb{R}^1$ represents the bias parameter, $*$ represents the convolution, σ represents ReLU, and $[\cdot]$ represents the concatenation by the channel. For convenience, the formula of $f(X_k^l)$ is simplified

$$f(X_k^l) = \sigma\left(W_f * X_k^l + b_f\right) \tag{3}$$

where W_f represents the weight parameters and b_f represents the bias parameter. The formulas of $g(X_k^l)$ and $h(X_k^l)$ are similar to the formula of $f(X_k^l)$

$$g(X_k^l) = \sigma\left(W_g * X_k^l + b_g\right) \tag{4}$$

$$h(X_k^l) = \sigma\left(W_h * X_k^l + b_h\right) \tag{5}$$

where W_g and W_h represent the weight parameters of convolution and b_g and b_h represent the bias parameters. Second, the sizes of $f(X_k^l)$, $g(X_k^l)$ and $h(X_k^l)$ are reshaped to $ss \times o$, where ss is calculated by multiplying the height and width

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of convolutional features. The relationship $R \in \mathbb{R}^{ss\times ss}$ of different pixels in the HSI cube X_k is calculated by

$$R = f\left(X_k^l\right) g\left(X_k^l\right)^T. \tag{6}$$

Third, a softmax is utilized to normalize the R by row

$$\hat{R}(i,j) = \frac{e^{R(i,j)}}{\sum_{j=1}^{ss} e^{R(i,j)}}. \tag{7}$$

The ith row of normalized $\hat{R}$ indicates the correlation of other pixels with the ith pixel of HSI cubes. Next, $h(X_k^l)$ is multiplied by the normalized $\hat{R}$ to generate the attention features $Att \in \mathbb{R}^{ss\times o}$

$$Att = \hat{R}h\left(X_k^l\right). \tag{8}$$

Then, a convolutional layer with c_1 $o\times 1\times 1$ kernels is utilized to convert Att to $Att' \in \mathbb{R}^{s\times s\times c_1}$. Finally, to facilitate the convergence of the proposed method, the attention feature Att' is added back to the input feature X_k^l with the skip connection

$$\hat{X}_k^l = Att' + X_k^l. \tag{9}$$

In the attention module, the spatial size $s \times s$ of each feature is set to 7×7. The number o of convolutional kernels is set to 64. The detailed settings of attention module are shown in Algorithm 1.

Algorithm 1 Details of the Attention Module

Input:
Features: $X_k^l \in \mathbb{R}^{s\times s\times c_1}$.
Output:
Features of attention module: $\hat{X}_k^l \in \mathbb{R}^{s\times s\times c_1}$.
Initialization:
Initialize all weight parameters of convolutional kernels with the *Xavier* method [53].
1: Convolve X_k^l with the $\{W_f, b_f\}$, $\{W_g, b_g\}$, and $\{W_h, b_h\}$ to generate the $f(X_k^l)$, $g(X_k^l)$, and $h(X_k^l) \in \mathbb{R}^{s\times s\times o}$;
2: Reshape the size of convolutional features $f(X_k^l)$, $g(X_k^l)$, and $h(X_k^l)$ to $ss \times o$;
3: Calculate the relationship $R \in \mathbb{R}^{ss\times ss}$ of each hyperspectral pixel by multiplying $f(X_k^l)$ by $g(X_k^l)^T$;
4: Normalize R with the softmax function to generate $\hat{R} \in \mathbb{R}^{ss\times ss}$;
5: Extract the attention features $Att \in \mathbb{R}^{ss\times o}$ by multiplying $\hat{R}$ by $h(X_k^l)$;
6: Reshape the size of attention features Att to $s \times s \times o$;
7: Convert Att to $\hat{Att} \in \mathbb{R}^{s\times s\times o}$ with a convolutional layer with c_1 $o \times 1 \times 1$ convolutional kernels;
8: Finally the attention features $\hat{Att}$ is added back to the spectral features X_k^l to obtain the $\hat{X}_k^l \in \mathbb{R}^{s\times s\times c_1}$.

D. SSAN

As mentioned earlier, the spectral module, the spatial module, and the attention module are composed of several simple 3-D convolutional layers. These modules can easily be integrated to form an end-to-end SSAN. After extracting the spectral–spatial features from HSI cubes, the fully connected layer with ReLU and the softmax classifier are employed for HSI classification. Finally, the classic cross entropy is exploited as the loss function

$$\text{Loss} = -\frac{1}{\text{bs}}\sum_{k=1}^{\text{bs}}\sum_{p=1}^{n} \mathbf{1}\{Y_k = p\} \log \frac{e^{\theta_p^T X_k^{\text{feat}}}}{\sum_{i=1}^{n} e^{\theta_i^T X_k^{\text{feat}}}} \tag{10}$$

where X_k^{feat} represents the spectral–spatial feature of the HSI cube X_k, Y_k represents the land-cover label of the HSI cube X_k, bs represents the number of batch size, n represents the number of land-cover categories, $\mathbf{1}\{Y_k = p\}$ represents the indicator function (if Y_k is the same as p, $\mathbf{1}\{Y_k = p\}$ is 1; else, $\mathbf{1}\{Y_k = p\}$ is 0). Furthermore, the batch normalization is employed for each layer and the dropout layer is added behind the fully connected layer to avoid overfitting.

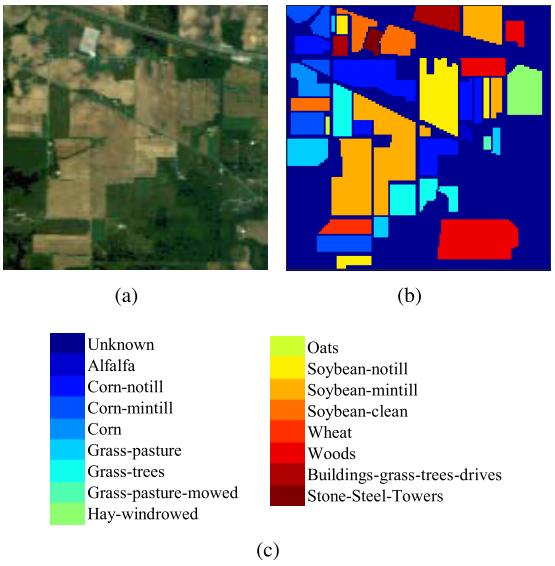


Fig. 6. Indian Pines database. (a) False color composite. (b) Ground truth. (c) Color code.

IV. Experiments

A. Databases and Evaluation Measures

The performance of SSAN is evaluated on three public remote sensing hyperspectral databases: Indian Pines, Pavia University, and Salinas.[1]

The Indian Pines database is collected by the AVIRIS sensor. This database contains 220 spectral bands with wavelengths ranging from 0.4×10^{-6} to 2.5×10^{-6} m. The spatial size of the Indian Pines is 145×145. On the Indian Pines, 10% of the labeled pixels in each land-cover category are randomly selected for training and the remaining for testing. The false-colour composite, ground truth, and color code are shown in Fig. 6.

The Pavia University database is gathered by the ROSIS sensor. This image contains 103 spectral bands with wavelengths ranging from 0.38×10^{-6} to 0.86×10^{-6} m. The spatial

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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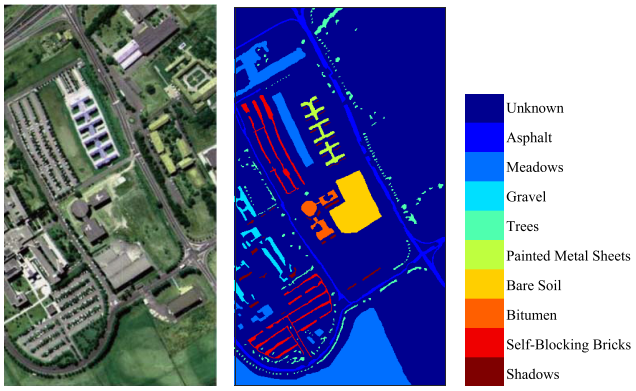


Fig. 7. Pavia University database. (a) False color composite. (b) Ground truth. (c) Color code.

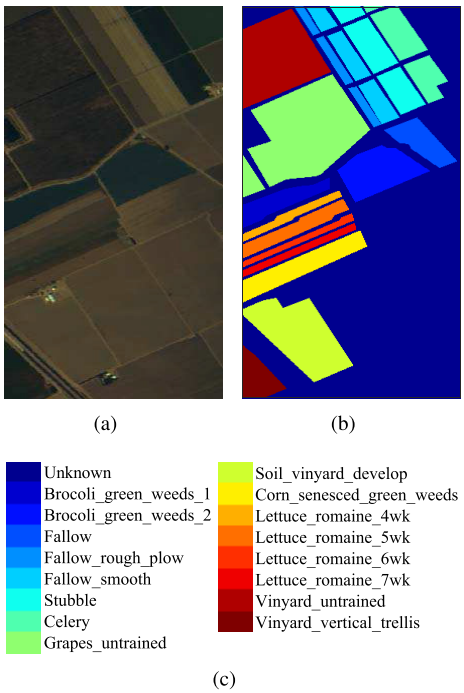


Fig. 8. Salinas database. (a) False color composite. (b) Ground truth. (c) Color code.

size of Pavia University is 610×240. On Pavia University, 2% of the labeled pixels in each land-cover category are randomly chosen as training samples and the rest as testing samples. The false-colour composite, ground truth, and color code are shown in Fig. 7.

Salinas database is acquired by the AVIRIS sensor. This database has 224 spectral bands with wavelengths ranging from 0.4×10^{-6} to 2.5×10^{-6} m. The spatial size of the Salinas is 512×217. On the Salinas, 2% of the labeled pixels in each land-cover category are randomly chosen for training. The false-colour composite, ground truth, and color code are shown in Fig. 8.

In order to quantify the classification performance of SSAN, the overall accuracy (OA), average accuracy (AA), kappa coefficient (kappa), and accuracy of each land-cover categories are employed as evaluation measures. OA is the ratio of the number of correctly labeled hyperspectral pixels to the total number of hyperspectral pixels in test samples. AA is the mean of accuracies in different land-cover categories. Kappa measures the consistency between classification results and ground truth. The larger values of OA, AA, and kappa represent the better classification results. Let $M \in \mathbb{R}^{n \times n}$ represent the error matrix of classification results, where n is the number of land-cover categories and the value of M in (i, j) position indicates the number of ith category samples that are classified to jth category. The values of OA, AA, and kappa can be calculated as follows:

$$\mathrm{OA} = \mathrm{sum}(\mathrm{diag}(M))/\mathrm{sum}(M)) \tag{11}$$

$$\mathrm{AA} = \mathrm{mean}(\mathrm{diag}(M)./\mathrm{sum}(M, 2)) \tag{12}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - (\mathrm{sum}(M, 1)\mathrm{sum}(M, 2))/\mathrm{sum}(M)^2}{1 - (\mathrm{sum}(M, 1)\mathrm{sum}(M, 2))/\mathrm{sum}(M)^2} \tag{13}$$

where $\mathrm{diag}(M) \in \mathbb{R}^{n \times 1}$ is a vector of diagonal elements of M, $\mathrm{sum}(\cdot) \in \mathbb{R}^1$ is the sum of all elements, $\mathrm{sum}(\cdot, 1) \in \mathbb{R}^{1 \times n}$ is the vector of the sum of elements in each column, $\mathrm{sum}(\cdot, 2) \in \mathbb{R}^{n \times 1}$ is the vector of the sum of elements in each row, $\mathrm{mean}(\cdot) \in \mathbb{R}^1$ is the mean of all elements, and $./$ represents the elementwise division. Finally, the visual classification results are also illustrated in this article.

B. Experimental Setting

In this section, the data preprocessing and the data augmentation are introduced.

In data preprocessing, first, the hyperspectral data are normalized with 14. Then, the normalized hyperspectral data are subtracted from the average value of the corresponding channel. Detailed formulations are as follows:

$$N(X(i, j, m)) = \frac{X(i, j, m) - \min(X)}{\max(X) - \min(X)} \tag{14}$$

$$\hat{X}(i, j, m) = N(X(i, j, m)) - \mathrm{mean}(N(X(:, :, m))) \tag{15}$$

where X represents the original hyperspectral data, $X(i, j, m)$ represents the value of X in the (i, j) spatial position and the mth spectral band, $N(X(i, j, m))$ represents the normalized $X(i, j, m)$, $\hat{X}(i, j, m)$ represents the preprocessed $X(i, j, m)$, $\min(X)$ represents the minimum of X, $\max(X)$ represents the maximum of X, and $\mathrm{mean}(N(X(:, :, m)))$ represents the mean of normalized X in the mth spectral band.

Specifically, the data augmentation is utilized to alleviate the problem of insufficient labeled hyperspectral pixels. First, the HSI cube is flipped horizontally and vertically. Second, the HSI cube is rotated 90°, 180°, and 270°. After data augmentation, extra five HSI cubes are generated from one HSI cube.

C. Number of Convolutional Layers

In this section, several experiments are conducted to explore the performance of the proposed SSAN with a different number of convolutional layers. The proposed SSAN contains a spectral module and a spatial module. The number of convolutional layers in each module has a large impact on

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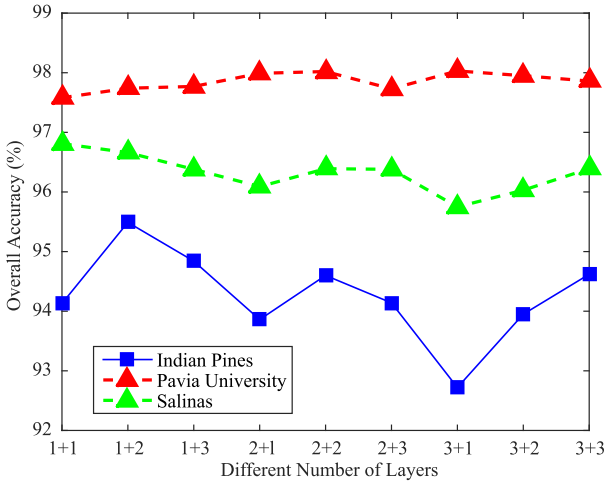


Fig. 9. OAs of SSAN with a different number of convolutional layers on the Indian Pines, Pavia University, and Salinas databases. "$M + N$" represents that the spectral module is combined with M convolutional layers and the spatial module is combined with N convolutional layers.

performance. We set the number of convolutional layers in the spectral module and the spatial module from 1 to 3. The OAs of SSAN with a different number of convolutional layers are shown in Fig. 9. In Fig. 9, "$M + N$" of the horizontal axis represents that the spectral module is combined with M convolutional layers and the spatial module is combined with N convolutional layers. As shown in Fig. 9, on the Indian Pines, the proposed SSAN with the setting of "1 + 2" shows superior performance. On Pavia University, the proposed SSAN with the setting of "2 + 2" shows superior performance. On the Salinas, the proposed SSAN with the setting of "1 + 1" shows superior performance. In order to easily understand the network structure of SSAN, the detailed structures of SSAN on each database are shown in Table I. The kernel number of each convolutional layer is also reported in Table I. The batch normalization and ReLU following each convolutional layer and the first FC layer are omitted. The second FC layer is not configured with the batch normalization and ReLU.

In the proposed SSAN, multiple attention modules are embedded in the network to extract attention features. Multiple attention modules are helpful to enhance the discrimination of spectral–spatial features. However, the attention module is a flexible module that can be embedded anywhere in SSAN. Several experiments are conducted to explore the performance of the proposed method with the attention modules embedded after different layers. The OAs are reported in Table II. In Table II, "spe" represents that the attention module is embedded after the spectral module, "spa1" represents that the attention module is embedded after the first convolutional layer of the spatial module, "spa2" represents that the attention module is embedded after the second convolutional layer of the spatial module, "+" represents that the attention modules are embedded after two layers (for example, "spe + spa1" represents that the attention modules are embedded after the spectral module and the first convolutional layer of the spatial module), "spe + spa1 + spa2" represents the proposed SSAN on the Indian Pines and Pavia University database, and

TABLE I

DETAILED STRUCTURES OF THE PROPOSED SSAN ON EACH DATABASE

Layers		Kernel number	Indian Pines	Pavia University	Salinas
Input		-	✓	✓	✓
Spectral module	3D-convolution	32	✓	✓	✓
	3D-convolution	32	-	✓	-
	Attention module	64	✓	✓	✓
Spatial module	3D-convolution	64	✓	✓	✓
	Attention module	64	✓	✓	✓
	3D-convolution	64	✓	✓	-
	Attention module	64	✓	✓	-
FC layer		256	✓	✓	✓
Dropout		0.5	✓	✓	✓
FC layer		n	✓	✓	✓
Softmax		-	✓	✓	✓

1. n represents the number of land-cover categories.
2. FC layer represents the fully connected layer.
3. '✓' represents that SSAN is configured with the current layer and '-' represents that SSAN is not configured with the current layer.

TABLE II

OAS (%) WITH DIFFERENT STYLES OF ATTENTION MODULE EMBEDDING

	Indian Pines	Pavia University	Salinas
without attention	94.65	97.37	96.23
spe	95.11	97.73	96.49
spa1	94.89	97.58	96.27
spa2	94.74	97.65	-
spe + spa1	95.14	97.85	**96.81**
spe + spa2	94.94	97.84	-
spa1 + spa2	94.78	97.82	-
spe + spa1 + spa2	**95.49**	**98.02**	-

"spe + spa1" represents the SSAN on the Salinas database. On the Salinas database, the spatial module of SSAN is composed of a convolutional layer. Therefore, on the Salinas database, there are only three kinds of attention module embedding: "spe," "spa1," and "spe + spa1." As shown in Table II, compared with the "without attention" kind, the proposed attention module can effectively improve OAs. The proposed SSAN with multiple attention modules has excellent performance.

D. Comparing With Other Methods

To verify the effectiveness of the proposed SSAN, the classification results of SSAN are compared with spectral–spatial DL-based methods: 2-D CNN [25], SMBN [29], DFFN [30], deformable HSI classification networks (DHCNet) [27], and spectral–spatial residual network (SSRN) [9]. In addition, the spectral module, the spatial module, and the SSN are also evaluated on these HSI databases. For a fair comparison, the proposed method and compared methods adopt the same experimental settings, including data preprocessing, data augmentation, and hyperparameter settings. The detailed hyperparameter settings are reported as follows. The spatial size of HSI cubes for all methods is set to 7×7. The number of batch size is set to 100. The number of training epoch is set to 200. The weight parameters of all methods are optimized with the Adam [54]. The learning rate of the proposed method is set to 0.01 and then divided by 2 after every 20 epochs. The initial learning rates of compared methods are set the same as values reported in original works. All HSI classification methods

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TABLE III
NUMBER OF TRAINING SAMPLES AND TESTING SAMPLES ON THE INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATABASES

Indian Pines				Pavia University				Salinas			
Class	Name	Training	Testing	Class	Name	Training	Testing	Class	Name	Training	Testing
1	Alfalfa	5	41	1	Asphalt	133	6498	1	Brocoli_g_w_1	41	1968
2	Corn-n	143	1285	2	Meadows	373	18276	2	Brocoli_g_w_2	75	3651
3	Corn-m	83	747	3	Gravel	42	2057	3	Fallow	40	1936
4	Corn	24	213	4	Trees	62	3002	4	Fallow_r_p	28	1366
5	Grass-p	49	434	5	Painted M S	27	1318	5	Fallow_s	54	2624
6	Grass-t	73	657	6	Bare-S	101	4928	6	Stubble	80	3879
7	Grass-p-m	3	25	7	Bitumen	27	1303	7	Celery	72	3507
8	Hay-w	48	430	8	Self-B B	74	3608	8	Grapes_u	226	11045
9	Oats	2	18	9	Shadows	19	928	9	Soil_v_d	125	6078
10	Soybean-n	98	874					10	Corn_s_g_w	66	3212
11	Soybean-m	246	2209					11	Lettuce_r_4wk	22	1046
12	Soybean-c	60	533					12	Lettuce_r_5wk	39	1888
13	Wheat	21	184					13	Lettuce_r_6wk	19	897
14	Woods	127	1138					14	Lettuce_r_7wk	22	1048
15	Buildings-g-t-d	39	347					15	Vinyard_u	146	7122
16	Stone-s-t	10	83					16	Vinyard_v_t	37	1770
	Total	1031	9218		Total	858	41918		Total	1092	53037

TABLE IV
CLASSIFICATION RESULTS (%) ON THE INDIAN PINES DATABASE

Class	2D-CNN [25]	SMBN [29]	DFFN [30]	DHCNet [27]	SSRN [9]	Spectral	Spatial	SSN	**SSAN**
1	78.05	82.93	**100**	82.93	82.93	56.10	70.73	80.49	80.49
2	79.77	82.80	91.44	91.44	**92.37**	74.55	82.96	91.21	90.82
3	83.94	92.77	93.44	88.76	**97.86**	83.53	84.47	93.44	93.84
4	90.14	84.51	**94.37**	83.10	62.44	48.36	64.79	69.95	89.20
5	**99.54**	97.93	97.24	96.31	97.47	96.77	96.77	98.39	99.08
6	98.63	99.70	99.24	99.24	99.85	**100**	99.85	99.70	99.24
7	76.00	96.00	68.00	84.00	60.00	56.00	**100**	92.00	96.00
8	99.77	96.51	99.77	97.21	**100**	99.30	**100**	98.84	98.14
9	**100**	**100**	88.89	88.89	44.44	**100**	94.44	94.44	**100**
10	89.24	89.36	93.82	92.68	91.30	77.00	90.62	90.85	**94.62**
11	95.74	97.69	95.88	96.15	97.69	92.49	96.42	**99.09**	98.10
12	76.55	85.74	84.80	86.30	91.37	59.66	71.48	90.43	**94.56**
13	**100**	94.57	97.83	95.65	**100**	98.91	99.46	95.11	**100**
14	**99.21**	97.19	98.42	97.28	98.59	96.84	97.45	98.59	98.42
15	85.30	82.42	**96.54**	88.47	82.71	69.16	82.71	82.71	82.71
16	**100**	95.18	98.80	92.77	**100**	96.39	96.39	92.77	91.57
OA	91.31	92.74	95.01	93.68	94.77	85.51	90.84	94.65	**95.49**
AA	90.74	92.21	93.65	91.32	87.44	81.57	89.28	91.75	**94.17**
kappa	90.06	91.70	94.31	92.78	94.03	83.35	89.51	93.88	**94.85**

TABLE V
CLASSIFICATION RESULTS (%) ON THE PAVIA UNIVERSITY DATABASE

Class	2D-CNN [25]	SMBN [29]	DFFN [30]	DHCNet [27]	SSRN [9]	Spectral	Spatial	SSN	**SSAN**
1	93.63	94.51	94.24	96.03	97.78	94.71	92.11	**98.71**	98.68
2	98.20	99.56	99.10	98.68	98.59	**99.93**	99.80	99.03	99.44
3	92.32	78.80	86.49	83.33	**95.77**	84.74	77.73	84.10	86.00
4	**99.00**	95.74	97.57	98.83	95.07	89.01	98.30	98.50	98.33
5	99.62	94.08	98.33	97.95	**100**	99.54	98.18	96.59	99.92
6	99.39	96.96	96.63	96.88	98.94	94.03	97.87	**99.78**	99.11
7	94.93	86.88	87.87	86.80	93.32	73.14	73.91	94.40	**96.55**
8	83.76	83.12	92.82	93.04	90.96	**94.40**	87.67	90.60	94.07
9	99.25	99.14	98.92	**100**	99.68	99.35	**100**	**100**	**100**
OA	96.12	95.19	96.41	96.47	97.36	95.57	95.29	97.37	**98.02**
AA	95.57	92.09	94.66	94.62	96.68	92.10	91.73	95.57	**96.90**
kappa	94.89	93.61	95.24	95.32	96.51	94.08	93.76	96.52	**97.37**

do not employ the PCA to reduce the spectral dimension of HSIs. On each database, a detailed number of training samples and testing samples for each land-cover category is shown in Table III.

The first experiment is conducted on the Indian Pines database. For all HSI classification methods, 10% of the labeled pixels in each land-cover category are randomly selected as training samples. The quantitative classification results of compared methods are shown in Table IV. The pixel number of different land-cover categories on the Indian Pines is different, resulting in the imbalance in the training samples of different land-cover categories. In the "Alfalfa,"

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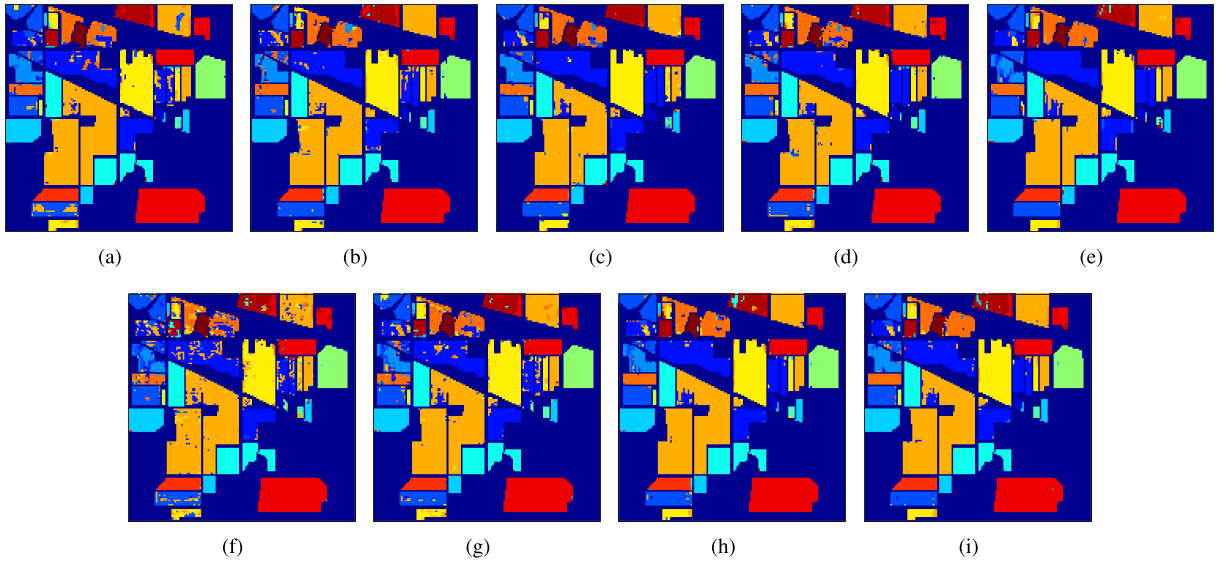


Fig. 10. Visual classification results on the Indian Pines. (a) 2-D CNN. (b) SMBN. (c) DFFN. (d) DHCNet. (e) SSRN. (f) Spectral module. (g) Spatial module. (h) SSN. (i) SSAN.

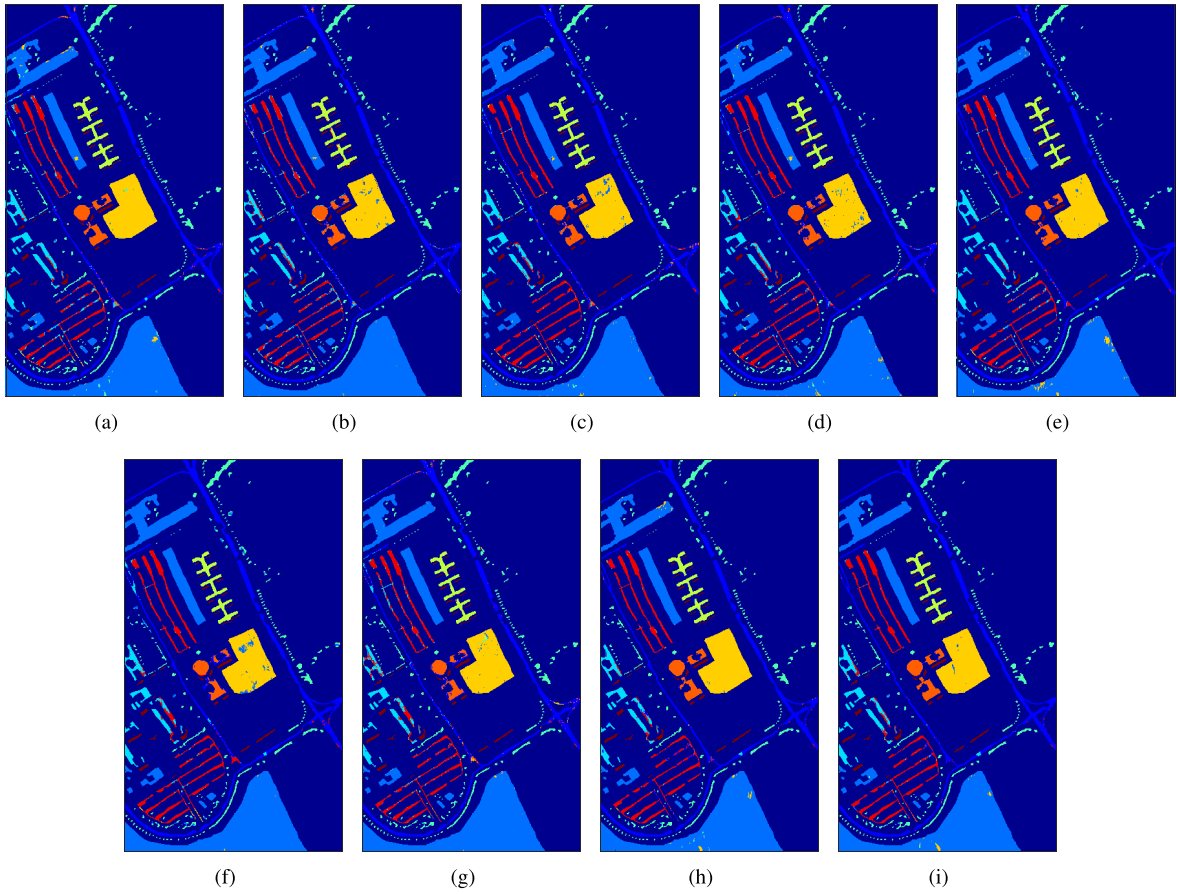


Fig. 11. Visual classification results on the Pavia University. (a) 2-D CNN. (b) SMBN. (c) DFFN. (d) DHCNet. (e) SSRN. (f) Spectral module. (g) Spatial module. (h) SSN. (i) SSAN.

"Grass-p-m," and "Oats" categories, only 5, 3, and 2 pixels are used as training samples, respectively. Insufficient training samples make the compared methods perform poorly in these land-cover categories. As shown in Table IV, the accuracies of 2-D CNN in the "Alfalfa" and "Grass-p-m" categories are 78.08% and 76.00%, respectively. DFFN achieves an accuracy of 68.00% in the "Grass-p-m" category. SSRN achieves an accuracy of 60.00% in the "Grass-p-m" category and an accuracy of 44.44% in the "Oats" category. Especially, the proposed SSAN shows superior performance in the "Alfalfa,"

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TABLE VI
CLASSIFICATION RESULTS (%) ON THE SALINAS DATABASE

Class	2D-CNN [25]	SMBN [29]	DFFN [30]	DHCNet [27]	SSRN [9]	Spectral	Spatial	SSN	**SSAN**
1	**100**	81.20	99.49	99.54	**100**	**100**	99.95	98.42	98.78
2	98.63	99.73	99.81	99.89	97.89	**99.97**	99.07	**99.97**	**99.97**
3	94.37	89.72	96.07	97.42	**99.48**	98.81	94.68	98.14	98.66
4	98.17	98.83	99.34	98.10	**99.63**	98.46	98.39	97.95	99.05
5	77.06	98.02	99.35	97.98	99.39	**99.47**	97.56	98.09	99.39
6	**100**	99.97	99.90	99.97	**100**	**100**	**100**	**100**	99.97
7	99.91	99.37	99.63	99.71	**100**	99.77	99.86	99.94	99.91
8	87.93	89.13	91.44	89.37	88.26	**93.69**	81.38	91.82	92.46
9	**100**	99.42	98.93	**100**	**100**	99.97	99.62	99.97	99.95
10	95.83	96.67	95.98	96.82	**97.70**	96.86	95.36	96.82	96.33
11	99.33	97.99	98.37	99.04	**99.90**	99.33	96.18	99.24	99.43
12	98.94	97.03	**100**	99.79	**100**	99.95	99.74	99.47	**100**
13	99.89	99.44	98.55	95.88	**100**	99.89	99.67	**100**	**100**
14	97.52	98.47	99.52	**100**	99.33	99.90	98.85	98.66	99.81
15	91.01	85.73	88.46	85.41	**93.30**	78.22	79.88	89.15	91.39
16	89.21	97.91	99.32	98.70	**99.15**	98.31	97.68	97.80	98.19
OA	94.08	94.03	95.96	95.21	96.27	95.37	92.47	96.23	**96.81**
AA	95.49	95.54	97.76	97.35	**98.38**	97.66	96.12	97.84	98.33
kappa	93.41	93.36	95.50	94.66	95.85	94.83	91.63	95.80	**96.54**

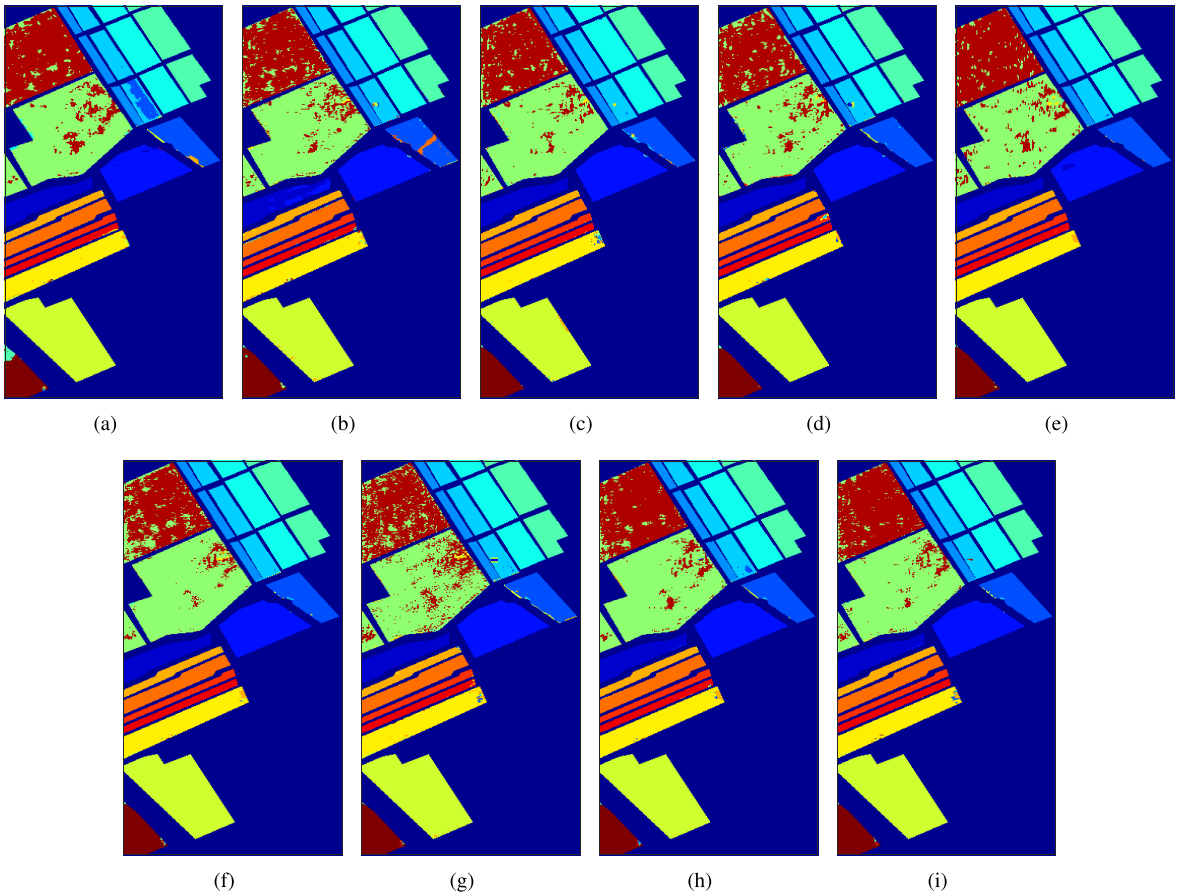


Fig. 12. Visual classification results on the Salinas. (a) 2-D CNN. (b) SMBN. (c) DFFN. (d) DHCNet. (e) SSRN. (f) Spectral module. (g) Spatial module. (h) SSN. (i) SSAN.

"Grass-p-m," and "Oats" categories, and the accuracies are 80.49%, 96.00%, and 100.00%, respectively. In terms of OA, AA, and kappa, the proposed SSAN outperforms the compared methods. By comparing the SSAN with the SSN, it can be seen that the attention module improves the OA of SSN by 0.84%, the AA of SSN by 2.42%, and the kappa of SSN by 0.97%. This indicates that the attention module can extract discriminative spectral–spatial features from attention areas and effectively mitigate the effects of interfering pixels. In addition, both the spectral module and the spatial module are vital in the proposed SSAN. The SSN, which consists of the spectral module and the spatial module, also shows the competitive performance.

The second experiment is conducted on the Pavia University database. For all HSI classification methods, 2% of the labeled pixels in each land-cover category are randomly

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TABLE VII
TRAINING TIME AND TESTING TIME (SECONDS) ON EACH DATABASE

	Methods	2D-CNN [25]	SMBN [29]	DFFN [30]	DHCNet [27]	SSRN [9]	Spectral	Spatial	SSN	SSAN
Indian	Training time	152.23	330.44	448.40	323.90	1943.50	966.08	378.15	1112.41	1783.42
	Testing time	0.46	0.79	0.70	1.42	3.37	1.62	1.08	2.49	3.82
Pavia	Training time	104.99	281.02	378.96	297.50	912.09	856.22	196.04	912.63	1266.23
	Testing time	1.33	2.61	3.67	5.72	7.66	6.82	2.56	8.71	11.81
Salinas	Training time	212.82	424.46	570.83	449.13	2360.56	1188.89	418.01	1389.40	2145.03
	Testing time	2.86	4.35	5.55	7.94	19.77	9.38	4.07	13.49	21.70

TABLE VIII
OAS (%) OF SSAN WITH DIFFERENT SPATIAL SIZES

	3×3	5×5	7×7	9×9	11×11
Indian Pines	86.93	93.83	95.49	95.62	96.13
Pavia University	95.89	97.57	98.02	98.11	97.65
Salinas	93.19	95.98	96.81	97.23	97.20

TABLE IX
OA (%) OF SSAN WITH DIFFERENT TRAINING RATIOS

Training Ratio	5%	10%	15%	20%
Indian Pines	88.71	95.49	96.54	97.94
Training Ratio	1%	2%	5%	10%
Pavia University	95.70	98.02	99.25	99.64
Training Ratio	1%	2%	5%	10%
Salinas	94.37	96.81	98.24	99.02

selected as training samples. The classification results are reported in Table V. The proposed SSAN outperforms several state-of-the-art methods on the Pavia University. As shown in Table III, compared with the Indian Pines database, the number of training samples for each land-cover category in the Pavia University database is relatively large. In the "Shadows" category of Pavia University database, the number of training samples is 19 more than the training sample number of "Oats" category on the Indian Pines database. With sufficient training samples, compared DL-based methods achieve superior performance. Even 2-D CNN achieves an OA of 96.12%. However, due to the influence of interfering pixels, compared methods cannot extract discriminative spectral–spatial features for classification. In this article, an attention module is employed to capture discriminative attention features. As shown in Table V, comparing SSAN with SSN, the attention module improves the OA of SSN by 0.65%, the AA of SSN by 1.33%, and the kappa of SSN by 0.85%. Comparing SSN with the spectral module and the spatial module, it can be seen that combining the spectral module and the spatial module can effectively extract spectral–spatial features.

The third experiment is performed on the Salinas database. Similar to the Pavia University database, 2% of the labeled pixels in each land-cover category are randomly selected as training samples and the rest samples are utilized as testing samples. The classification results are shown in Table VI. DL-based methods also achieve superior performance. In particular, 2-D CNN achieves an accuracy of 100% in the "Brocoli_g_w_1," "Stubble," and "Soil_v_d" categories. SSRN achieves an accuracy of 100% in the "Brocoli_g_w_1," "Stubble," "Celery," "Soil_v_d," "Lettuce_r_5wk," and "Lettuce_r_6wk" categories, respectively. In the "Grapes_u" category, the number of test samples is 11 045, far more than the number of test samples in other categories. The proposed SSAN achieves an accuracy of 92.46% in the "Grapes_u" category, which makes the proposed SSAN superior to compared methods in terms of OA. In terms of AA, the accuracy of SSRN is 0.05% higher than that of the proposed SSAN. As shown in Table VI, the attention module improves the OA of the SSN by 0.58%, the AA of the SSN by 0.49%, and the kappa of the SSN by 0.74%.

The visual classification results of different classification methods on each database are shown in Figs. 10–12. Comparing visual results with the ground truth, the proposed SSAN has a superior visual result on three public HSI databases. On the Indian Pines database, the land-cover areas are dense, which makes the HSI classification task more difficult. The proposed SSAN is superior to compared methods on the Indian Pines database, especially at the edges of land-cover areas. The visual results indicate that the attention module can effectively suppress the influence of interfering pixels. On the Pavia University and Salinas databases, due to the spectral variability caused by atmospheric effects and instrument noises, the visual results of compared methods are not smooth. The proposed SSAN can effectively alleviate the influence of spectral variability and achieves smooth visual results.

The training and testing time can intuitively show the computational efficiency of HSI classification methods. The training and testing time of the proposed SSAN and compared methods are reported in Table VII. In this article, all experiments are done with the Ubuntu operating system, Tensorflow-1.4.1 toolbox,[2] Intel Core i7-5930K, 64-GB RAM and GeForce GTX Titan X. The spectral module and the attention module employ a large number of convolutional kernels with the spatial size of 1×1, whose computational efficiency is low. Therefore, it takes a lot of time to train the proposed SSAN.

E. Impact of Spatial Size

The spatial size of cropped HSI cubes has a great influence on classification results [27], [30]. In this section, several experiments are conducted to explore the effects of different spatial sizes on the Indian Pains, Pavia University, and Salinas databases. The training ratios on the Indian Pains, Pavia University, and Salinas databases are 10%, 2%, and 2%, respectively. The spatial size of cropped HSI cubes is set from 3×3 to 11×11. The OAs of SSAN with different sizes of

[2]https://tensorflow.google.cn/

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HSI cubes are shown in Table VIII. When the spatial size of cropped HSI cubes becomes larger, cropped HSI cubes contain more spatial information. Therefore, a proper spatial size can improve the OA of classification. However, classification results do not always improve as the spatial size increases. Too large spatial size will also introduce more interference pixels, which affect the classification results. When the space size is 11×11, the OA of SSAN begins to decline on the Pavia University and Salinas databases.

F. Impact of Training Ratio

In this section, the performance of the proposed SSAN with different training ratios is explored. The spatial size of cropped HSI cubes is set to 7×7. On the Indian pines database, 5%, 10%, 15%, and 20% of the labeled pixels in each land-cover category are randomly selected as training samples. On the Pavia University and Salinas databases, 1%, 2%, 5%, and 10% of the labeled pixels in each land-cover category are randomly selected as training samples. The OAs are shown in Table IX. More training samples can significantly improve the performance of the proposed SSAN.

V. Conclusion

In this article, the attention mechanism is employed to build an end-to-end SSAN for HSI classification. First, a simple SSN is built to extract spectral–spatial features from HSI cubes. Then, the attention module is embedded into the SSN to obtain the SSAN. The proposed SSAN attempts to capture the discriminative spectral–spatial features from attention areas of HSI cubes for HSI classification. The experimental results demonstrate the effectiveness of the proposed SSAN. However, the attention module in the proposed SSAN is computationally inefficient, which makes the training time of SSAN much longer than that of SSN. In the future, we will explore an efficient attention mechanism for HSI classification.

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